

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Key idea : For low-speed electrons , the kinetic energy relation is $K \hat{=} (1/2) m v^2$, so de Broglie wavelength $\hat{=} h / p = h / (m v) \hat{=} (h / m) \hat{=} (2 K / m)$.

- Electron rest mass $m \hat{=} 9.11 \times 10^{-28}$ g
- 1 eV $\hat{=} 4.135 \times 10^{-12}$ erg
- Kinetic energy $K = 1$ eV $= 4.135 \times 10^{-12}$ erg
- 1 eV corresponds to frequency $\hat{=} E/h = (4.135 \times 10^{-12} \text{ erg}) / (6.625 \times 10^{-27} \text{ erg}\cdot\text{s}) \hat{=} 6.24 \times 10^{14} \text{ s}^{-1}$
- Wavelength $\hat{=} c / \hat{=} 3.00 \times 10^{10} \text{ cm/s} / 6.24 \times 10^{14} \text{ s}^{-1} \hat{=} 4.81 \times 10^{-5} \text{ cm}$
- Alternatively , de Broglie : $h/\hat{=} = p = \hat{=} (2 m K)$, giving $\hat{=} = h / \hat{=} (2 m K)$; using $K = 4.135 \times 10^{-12}$ erg and $m = 9.11 \times 10^{-28}$ g yields $\hat{=} \hat{=} 4.14 \times 10^{-10} \text{ m} = 4.14 \times 10^{-5} \text{ cm}$.

Thus , the de Broglie wavelength is about 4.14×10^{-5} cm , which corresponds to choice D .
</think>
<answer>
D. 3.14×10^{-5} cm
</answer>

